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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Machine Learning (course)

Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 0 ()

Week 1 ()

Week 2 ()

Week 3 ()

Week 4 ()

Week 5 ()

☐ Artificial Neural Networks I - Early Models (unit? unit=60&lesson=61)

☐ Artificial Neural

Week 5: Assignment 5 (Non Graded)

Assignment not submitted

Note : This assignment is only for practice purpose and it will not be counted towards the Final score.

1) Given N samples $x_1, x_2, \dots, x_N$ drawn independently from a Gaussian distribution with variance σ^2 and unknown mean μ , find the MLE of the mean.

1 point

☐

$$\mu_{MLE} = \frac{\sum_{i=1}^N x_i}{\sigma^2}$$

☐

$$\mu_{MLE} = \frac{\sum_{i=1}^N x_i}{2\sigma^2 N}$$

☐

$$\mu_{MLE} = \frac{\sum_{i=1}^N x_i}{N}$$

☐

$$\mu_{MLE} = \frac{\sum_{i=1}^N x_i}{N-1}$$

2) Continuing with the above question, assume that the prior distribution of the mean is also a Gaussian distribution, but with parameters mean μ_p and variance σ_p^2 . Find the MAP estimate of the mean.

1 point

☐

$$\mu_{MAP} = \frac{\sigma^2 \mu_p + \sigma_p^2 \sum_{i=1}^N x_i}{\sigma^2 + N \sigma_p^2}$$

☐

$$\mu_{MAP} = \frac{\sigma^2 + \sigma_p^2 \sum_{i=1}^N x_i}{\sigma^2 + \sigma_p^2}$$

Networks II -
Backpropagati
on (unit?
unit=60&lesso
n=62)

Artificial
Neural
Networks III -
Backpropagati
on Continued
(unit?
unit=60&lesso
n=63)

Artificial
Neural
Networks IV -
Training,
Initialization
and Validation
(unit?
unit=60&lesso
n=64)

Parameter
Estimation I -
The Maximum
Likelihood
Estimate (unit?
unit=60&lesso
n=65)

Parameter
Estimation II -
Priors and the
MAP estimate
(unit?
unit=60&lesso
n=66)

Parameter
Estimation III
(unit?
unit=60&lesso
n=67)

Practice:
Week 5:
Assignment 5
(Non Graded)
(assessment?
name=268)

Quiz: Week 5:
Assignment 5
(assessment?
name=297)



$$\mu_{MAP} = \frac{\sigma^2 + \sigma_p^2 \sum_{i=1}^N x_i}{\sigma^2 + N\sigma_p^2}$$



$$\mu_{MAP} = \frac{\sigma^2 \mu_p + \sigma_p^2 \sum_{i=1}^N x_i}{N(\sigma^2 + \sigma_p^2)}$$

3) Which among the following statements is (are) true?

1 point

- ☐ MAP estimates suffer more from overfitting than maximum likelihood estimates.
- ☐ MAP estimates are equivalent to the ML estimates when the prior used in the MAP is a uniform prior over the parameter space.
- ☐ One drawback of maximum likelihood estimation is that in some scenarios (hint: multinomial distribution), it may return probability estimates of zero.
- ☐ The parameters which minimize the expected Bayesian L1 Loss is the median of the posterior distribution.

4) Which of the following statements is/are true about Neural Networks?

1 point

- ☐ Neural Networks can model arbitrarily complex decision boundaries.
- ☐ Neural Networks can be used to emulate a Gaussian kernel SVM.
- ☐ Training of a neural network is very sensitive to the initial weights.
- ☐ Ideal initialization for weights would be setting all of them to zeros

5) Which of the following is/are true about the Perceptron classifier?

1 point

- ☐ It can learn a OR function
- ☐ It can learn a AND function
- ☐ The obtained separating hyperplane depends on the order in which the points are presented in the training process.
- ☐ For a linearly separable problem, there exists some initialization of the weights which might lead to non-convergent cases.

Check Answers and Submit

☐ Week 5
Feedback
Form :
Introduction To
Machine
Learning (unit?
unit=60&lesso
n=286)

☒ Week 5:
Solution (unit?
unit=60&lesso
n=215)

Week 6 ()

Week 7 ()

Week 8 ()

Week 9 ()

Week 10 ()

Week 11 ()

Week 12 ()

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()**

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Solving
Session -
July 2024 ()**

